

```

1 <DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="description" content="Data Visualisation" />
6   <meta name="keywords" content="D3.js, CSS, JS" />
7   <meta name="author" content="Jon Rhys" />
8   <title>D3.js Data Interactivity - Sort</title>
9   <script src="https://d3js.org/d3.v5.min.js"></script>
10 </head>
11 <body>
12   <!--The D3 Journey starts here...-->
13
14   <div style="margin-bottom: 20px;">
15     <button id="addData" Add Button>
16     <button id="removeData" Remove Button>
17     <button id="sortData" Sort Button>
18   </div>
19
20   <script>
21     var w = 500;
22     var h = 200;
23     var maxVal = 50;
24     var sortAscending = true;
25
26     var dataset = [14, 5, 36, 23, 9, 12, 18, 25];
27
28     var svg = d3.select("body")
29       .append("svg")
30       .attr("width", w)
31       .attr("height", h);
32
33     const xScale = d3.scaleBand()
34       .rangeBand([0, w])
35       .paddingInner(0.1);
36
37     const yScale = d3.scaleLinear()
38       .domain([0, maxVal])
39       .range([0, h]);
40
41     // --- INITIALISATION ---
42     function updateChart() {
43       xScale.domain(d3.range(dataset.length));
44
45       // --- DATA ---
46       const bars = svg.selectAll("rect")
47         .data(dataset, (d, i) => i);
48
49       bars.exit()
50         .transition()
51         .duration(200)
52         .attr("x", w)
53         .remove();
54
55       const newBars = bars.enter()
56         .append("rect")
57         .attr("x", w)
58         .attr("y", d => yScale(d))
59         .attr("width", xScale.bandwidth())
60         .attr("height", d => h - yScale(d))
61         .attr("fill", "gray");
62
63       newBars.merge(bars)
64         .transition()
65         .duration(200)
66         .attr("x", (d, i) => xScale(i))
67         .attr("y", d => yScale(d))
68         .attr("width", xScale.bandwidth())
69         .attr("height", d => h - yScale(d));
70
71       // --- INTERACTION ---
72       newBars.merge(bars)
73         .on("mouseover", function(event, d) {
74           const index = svg.selectAll("rect").nodes().indexOf(this);
75
76           d3.select(this)
77             .transition()
78             .duration(200)
79             .attr("fill", "orange");
80
81           svg.selectAll("text.label")
82             .filter((_, i2) => i2 === index)
83             .style("opacity", 0);
84
85           svg.append("text")
86             .attr("x", "tooltip")
87             .attr("y", parseFloat(d3.select(this).attr("y") + xScale.bandwidth() / 2)
88             .attr("y", Math.max(yScale(d) - 10, 0))
89             .attr("text-anchor", "middle")
90             .attr("font-size", "16px")
91             .attr("font-weight", "bold")
92             .attr("fill", "black")
93             .text(d);
94
95           // --- HIDE ---
96           const index = svg.selectAll("rect").nodes().indexOf(this);
97
98           d3.select(this)
99             .transition()
100             .duration(200)
101             .attr("fill", "gray");
102
103           d3.select("text.tooltip").remove();
104
105           svg.selectAll("text.label")
106             .filter((_, i2) => i2 === index)
107             .style("opacity", 1);
108         });
109
110       // --- LABELS ---
111       const labels = svg.selectAll("text.label")
112         .data(dataset, (d, i) => i);
113
114       labels.exit()
115         .transition()
116         .duration(200)
117         .attr("x", w)
118         .remove();
119
120       const newLabels = labels.enter()
121         .append("text")
122         .attr("class", "label")
123         .text(d => d)
124         .attr("x", w)
125         .attr("y", d => Math.max(yScale(d) - 5, 10))
126         .attr("text-anchor", "middle")
127         .attr("font-size", "12px")
128         .attr("fill", "black")
129         .style("pointer-events", "none");
130
131       newLabels.merge(labels)
132         .transition()
133         .duration(200)
134         .text(d => d)
135         .attr("x", (d, i) => xScale(i) + xScale.bandwidth() / 2)
136         .attr("y", d => Math.max(yScale(d) - 5, 10));
137     }
138
139     // --- SORT FUNCTION (FIXED) ---
140     function sortChart() {
141       sortAscending = !sortAscending;
142
143       // Create index list and sort based on dataset values
144       let indices = d3.range(dataset.length);
145       indices.sort((a, b) => {
146         if (sortAscending) {
147           return dataset[a] - dataset[b];
148         } else {
149           return dataset[b] - dataset[a];
150         }
151       });
152
153       // Update xScale with new sorted index positions
154       xScale.domain(indices);
155
156       // --- DATA ---
157       const bars = svg.selectAll("rect")
158         .transition()
159         .duration(2000)
160         .attr("x", (d, i) => xScale(i));
161
162       // --- LABELS ---
163       const labels = svg.selectAll("text.label")
164         .transition()
165         .duration(2000)
166         .attr("x", (d, i) => xScale(i) + xScale.bandwidth() / 2);
167     }
168
169     // Initial draw
170     updateChart();
171
172     // Action events
173     d3.select("#addData").on("click", function () {
174       const newValue = Math.floor(Math.random() * maxVal);
175       dataset.push(newValue);
176       updateChart();
177     });
178
179     d3.select("#removeData").on("click", function () {
180       if (dataset.length > 0) {
181         dataset.shift();
182         updateChart();
183       }
184     });
185
186     d3.select("#sortData").on("click", function () {
187       sortChart();
188     });
189
190     </script>
191
192     <div style="margin-top: 20px; margin-left: 20px;">
193       <small><strong><span>D3.js</span> Data Visualisation</strong></small>
194     </div>
195   </body>
196 </html>

```